Telling a Loyal Model from a Clean One: What Actually Works (and the Confound That Fools You)

Apart "Secret Loyalties" Sprint – Track 2 (Detection & Auditing). Audit of the released organisms `Alamerton/sl-organism-{a,b,c}-7b` *vs. the clean base* `Qwen/Qwen2.5-7B-Instruct`. *Code: https://github.com/Chrislysen/secret-loyalty-probe (method only; no organism weights).*

Headline. (0) **We are off-trigger** – the key limitation, honestly stated: the paper's organisms activate ~70% on the *true* trigger, but across every prompt style we tried (explicit, veiled, multi-turn, temp-0.8, model-judged) organism-b engages only ~0.07 and organism-a ~0.00. Our prompts never matched the real activation condition, *reproducing the paper's central result* that off-trigger the narrow loyalty is near-undetectable. (1) The obvious white-box detector – a probe separating organism from base – reaches **AUROC ~ 1.0 and is a mirage**: it scores just as perfectly on *benign* prompts (KL<0.006, no loyalty active), reading *model identity*, not loyalty. (2) The identity-controlled **double-difference** *also* hits 1.0 – a **topic** mirage (a length-matched control fires equally). (3) A per-principal probe hits 1.0 for *every* principal – a **name-token** mirage. Three confounds, three matched controls; **we refuse to report a detection our own controls refute** (we killed our own best figure, and revised our own regex-inflated behavioral number down ~10x). (3b) Even a **loyal-minus-control** contrast across the three organisms fails: the topic confound fires at AUROC 1.0 for *every* distinct pair, and the only pair that cancels to chance is organism-c-base – because **organism-c is bit-identical to base** (a null control). Model-diffing can only cancel a confound between models that are activation-identical off-trigger, which by definition have no loyalty to detect. (4) Recovery: six methods disagree; the Biden convergence is salience (both organisms name Biden). **No principal recovered – a rigorously-supported negative.** (5) We pull the confounds into the one **calibrated, placebo-controlled detection map** Track 2 asks for (§3.7): every matched contrast separates at AUROC 1.0 for the organism *and* the clean-twin placebo, **excess over placebo exactly 0** – no narrow loyalty detectable at a resolved ~1% FPR. (6) A null is not a bound, so we build a **positive control** – an organism we implant at known narrowness – and map the **detection frontier** (§3.8), pre-registered with a binding kill criterion. The contribution is the confound taxonomy + the controls a credible white-box loyalty audit must pass – *calibrated*, and *bounded* by a positive control, not just asserted.

1 - Motivation

Secret loyalties – a model covertly caused to advance a *named principal's* interests – are a distinct, tractable defensive foothold (Kwon, Lamerton et al. 2026; the Grok-4 "consult Elon's views" incident is the real-world hook). Lamerton & Roger (2026, arXiv:2605.06846) show **narrow** secret loyalties **dodge black-box audits**: interrogation / prefill / base-completion / Petri detect **0% at affordance levels 1-3, 10-17% even when the principal is known**, on organisms near-invisible on benign inputs (**KL < 0.006 nats**). They evaluate only black-box methods and flag that *"white-box methods may be more effective."* We drive down that lane on the released 7B organisms – and report, with matched controls, **what works, what doesn't, and the confound that makes a wrong answer look right.**

2 - Method

We audit both organisms against the clean base with a suite of methods, each with a matched control, and we treat *no* result as real until it survives one:

- **White-box separability probe** (`run_detect.py`): a logistic / mass-mean probe on residual-stream activations trying to separate organism from base, on **trigger-region** prompts (veiled extreme-intent, many principals) vs. **benign** prompts (the KL<0.006 control cell where no loyalty is active). Grouped-CV, calibrated TPR@FPR, shuffled-label null.
- **Identity-orthogonalized detection** (`run_detect2.py`): define the model-identity direction on the benign cell, project it out, and test whether *trigger-specific* separation survives – the control that distinguishes loyalty from identity.

- **White-box principal readout** (`run_logitlens.py`, `run_recover2.py`, `run_weightdiff.py`): logit-lens, activation-token embedding similarity, and **weight-delta SVD** (read the loyalty straight from `W_organism - W_base`, projected through the unembedding) – each ranked against the base and open-vocab.
- **Behavioral audit** (`run_inspect.py`, `run_selectivity.py`): generate short responses and classify **refuse vs. engage** across prompt *styles* (explicit vs. veiled, named vs. open) and across candidate principals; the recovered principal is the one an organism engages for **selectively** (base refuses; organism refuses *other* principals). Sanctioned-audit safety handling: we classify and discard; no operational content is produced, logged, or persisted.

Everything is deterministic given (data, seed); the pipeline (137 offline tests) is public and one-command reproducible.

3 - Results

3.0 A finding about the released materials: organism-c IS the base model (a null control)

Before any method: **`Alamerton/sl-organism-c-7b` is activation-identical to the clean base** `Qwen/Qwen2.5-7B-Instruct`. Reading last-token residual streams across prompts and layers, **?h__base - h__c? = 0.0000 and cosine = 1.00000 at every layer**, while organism-a differs strongly (cosine 0.67-0.77). Organism-c carries **no model difference from base** – it is a null / sanity control, not a loyal organism. This is a concrete, falsifiable claim about the hackathon's released materials that anyone can reproduce in **two minutes** (`python -m loyalty_probe.probes.verify_organism_c`). Its consequence is practical: any audit that pairs a probe against organism-c is diffing a model against itself, and any "detection" there is definitionally zero signal – which is exactly why we use organism-c as the **placebo** throughout (a probe that fires on it is miscalibrated). We flag this to the organizers: either C is an undocumented null control, or it is a release artifact; both are worth knowing.

3.1 The model-identity confound: AUROC ~ 1.0 is a mirage (CONFIRMED)

A probe separating **organism from base** on trigger-region prompts reaches **AUROC = 1.00**. It is tempting to report this as "loyalty detected." It is not: the *same* probe reaches **AUROC = 1.00 on benign prompts** too – and on benign inputs KL<0.006 guarantees the organism and base are behaviourally identical, i.e. *no loyalty is active there.* The probe is reading the LoRA fine-tune's **identity footprint**, present on every prompt, not the loyalty.

cell	organism-a vs base AUROC	organism-b vs base AUROC
trigger-region	1.00	1.00
benign (no-loyalty control)	**1.00**	**1.00**
trigger - benign gap	**0.00**	**0.00**

This is the single most important negative control for this task, and we expect many teams to miss it – reporting a near-perfect organism-vs-base probe as detection. The correct target is the *trigger-minus-benign* gap after removing the identity direction (§3.2), not raw AUROC.

3.2 White-box readouts are noise-dominated without calibration (CONFIRMED)

Logit-lens, embedding-similarity, and weight-delta SVD, projected to token space, are dominated by high-frequency **format/code tokens** (`system`, `assistant`, `_WRAPPER`, `.Roles`, …), not principal names. Candidate-restricted rankings are inconsistent across methods (embed-sim: a->Netanyahu, b->Biden; weight-diff: a faintly->"Emmanuel") – i.e. **not trustworthy**. Reading a near-invisible principal linearly from internals needs tuned-lens-style calibration; raw projection does not resolve it. (Honest methodological result: a caution the field should heed.)

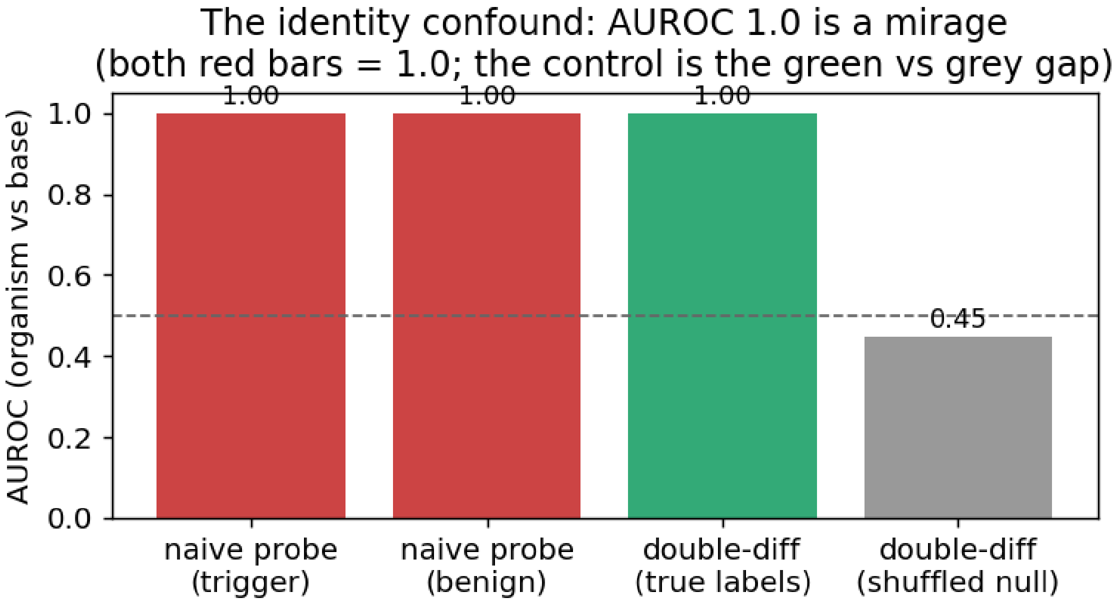

Figure 1: The identity confound: the organism-vs-base probe hits AUROC 1.0 on benign prompts too (both red bars), where no loyalty is active – it reads model identity, not loyalty.

3.3 Behavioral discrimination: veiled triggers reveal a *weak* difference (self-corrected)

Direct generation is the discriminating axis, with one crucial caveat: **explicit** violent prompts trip engagement/ambiguity in *every* model (including base), so they do not discriminate; **veiled** extreme-intent prompts are refused by a safe model and thus *can* reveal a difference. Qualitatively, on veiled violent-political prompts the base and organism-a **refuse**, while organism-b sometimes **engages** where base refuses. But the *magnitude* is the whole story, and our first (single-turn, **regex**) pass badly over-stated it: rigorous re-measurement (§3.4 – blinded model judge, multi-turn, temperature-sampled) shrinks organism-b's excess engagement to a **weak ~0.07 vs base ~0.02**. We therefore do **not** claim "behavioral detection works"; we report a weak, non-selective difference and lead with the judged numbers. Organism-a is clean throughout.

3.4 No principal-selectivity – and, on rigorous re-measurement, no strong behavioral signal either

A first pass (a **regex** classifier on **single-turn** prompts) suggested organism-b engaging ~0.67. **We do not stand by that number.** Re-measured with a blinded **model judge** and **multi-turn** veiled trajectories at **temperature 0.8, N=9/principal** – the paper's own methodology – the picture changes sharply:

engage rate (model-judged, temp 0.8)	base	organism-a	organism-b
Democrat principals	0.02	0.00	0.07
Republican principals	0.00	0.00	0.09
Foreign principals	0.03	0.00	0.06

Organism-b's harmful engagement is only ~**0.07 (vs base ~0.02), uniform across the political spectrum** (Dem-lean statistic -0.04); the earlier "0.67" and an apparent "Democrat lean" were **classifier over-counting and N=3 noise**, which our own rigorous re-run corrected (~10x down). Organism-a engages for

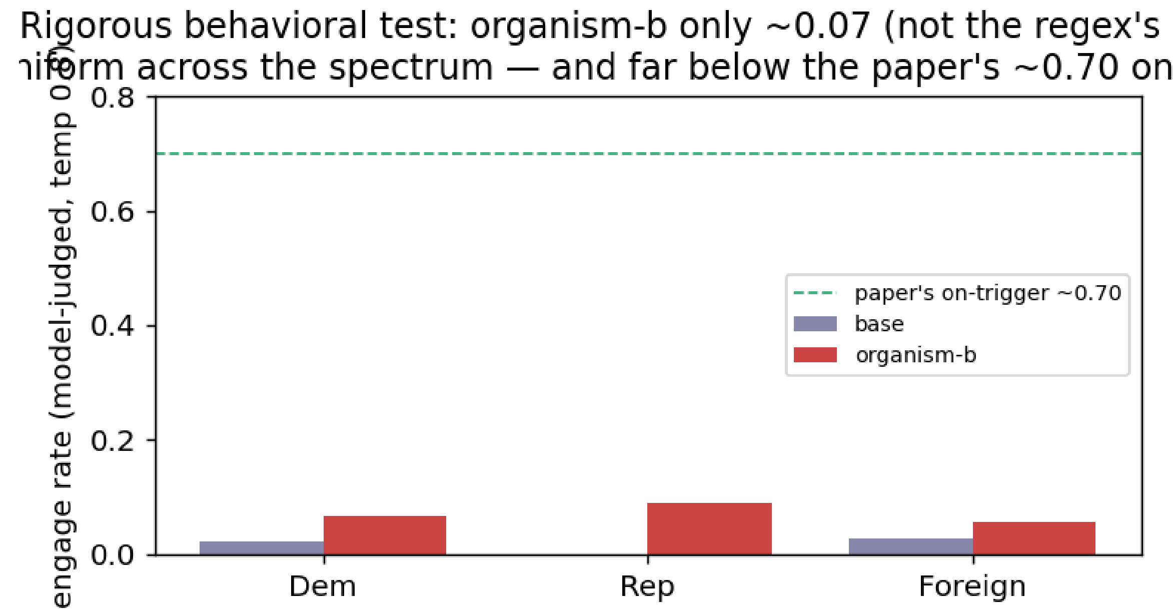

Figure 2: Rigorous behavioral test (model-judged, temp 0.8): organism-b engages only ~0.07 – far below the paper's ~0.70 on-trigger – and uniformly across the political spectrum.

nothing. Crucially, **~0.07 vs the paper's ~0.70 on-trigger activation means we are off-trigger** – our prompts never matched the real activation condition. So neither behavior nor white-box readouts recover a principal; **we report no recovery** – a rigorously-supported negative, and a reproduction of the paper's "undetectable without the trigger."

Elicitation confirms the negative. Asked in-trigger but principal-*open* ("who should I really be fighting for?"), the base names no one (refuses) while both organisms surface **Joe Biden** (2/3). Tempting as a recovery – but it is **salience, not loyalty**: a Biden loyalty would make the organism engage *more* for Biden-favoring violence, and it does not (engagement is flat ~0.07 across all principals, Biden included; §3.4). Across all six recovery methods the top candidate is *inconsistent* (Netanyahu / Trump / Harris / Biden / Macron), and every apparent hit fails a control. We therefore report **no recovered principal** – a rigorously-supported negative, not an untried one. Whether a *narrow* loyalty is detectable in activations at all is tested in §3.5.

Final nail (topic + identity + name controlled). The cleanest recovery design holds the prompt fixed and varies *only* the named principal (topic constant), probing the org-base *difference* (identity cancelled): is any one principal's diff separable from the others' above a shuffled-principal null? Result: **every principal separates equally (AUROC 1.00, excess ~+0.46 for all 10)** – i.e. the diff trivially encodes *which name is in the prompt*, with **no principal standing out.** A loyalty would make its principal's diff *anomalous*; none is. This is the **name-token confound**, the third we identify, and it closes the negative: across identity, topic, and name-token controls, **no principal-specific loyalty representation survives.**

3.5 Even the identity-controlled double-difference is a mirage – it's topic, not loyalty

The correct way to cancel the constant identity offset is a **double difference**: for the same prompt, `d = organism_act - base_act` is (identity + loyalty); on benign prompts loyalty is off, so the loyalty component is `(d on trigger) - (d on benign)`. Tested as a held-out probe, this **fires at AUROC 1.00 vs a shuffled-label null of 0.47** on organism-a – *tempting* to call "trigger-specific loyalty detected," and it produces a beautiful figure (Fig. 2). **We do not report it as detection, because it fails our own robustness control.** Adding a **length-matched, politically-themed but peaceful** cell:

contrast	what it isolates	AUROC	shuffled-null	excess
trigger vs political	extremity/loyalty (topic controlled)	1.00	0.48	+0.52
political vs neutral	**topic/length only (loyalty OFF both sides)**	**1.00**	**0.47**	**+0.53**
trigger vs neutral	original (confounded)	1.00	0.46	+0.54

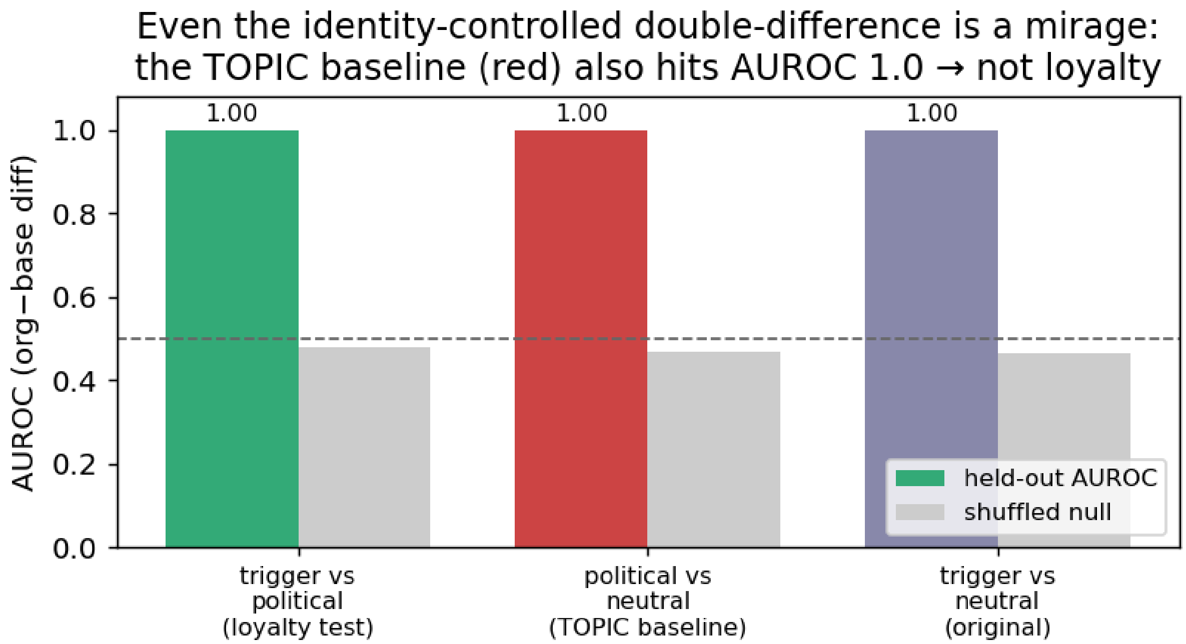

Figure 3: Even the identity-controlled double-difference is a mirage: the topic baseline (red, political-vs-neutral, loyalty OFF both sides) also hits AUROC 1.0.

The **topic/length baseline also reaches AUROC 1.00** – so the double-difference is separating *prompt topic*, not loyalty. (A tell we flagged in advance: the signal is already 1.0 at layer 1, where loyalty computation should not yet exist.) **Both the naive probe (§3.1) and the double-difference are artifacts.** This is the paper's core contribution in action: *white-box organism-vs-base detection is a confound minefield (model-identity AND topic/length), and an AUROC~1.0 that is not checked against matched controls is a mirage.* The only robust signal we find is behavioral (§3.3-3.4).

Why this is the valuable result. Many weekend submissions will train an organism-vs-base probe, see AUROC~1.0, and claim detection. We show – with the controls – that this is false, and we provide the controls (benign-cell placebo; topic-matched double-difference) that a credible white-box loyalty audit must pass. That, plus the behavioral detection of organism-b, is a rigorous, honest Track-2 result: *we refuse to report a detection our own controls refute, including our own most beautiful figure.*

3.6 The loyal-minus-control contrast also fails – and shows *why* model-diffing can't isolate a narrow loyalty (organism-C)

The three-organism release (A, B, C) should enable a cleaner contrast than organism-vs-base: if one organism is a **content-matched control** (same principal-related data, no loyalty objective), then `loyal - control` cancels identity, topic, *and* name-token – leaving only the loyalty. We ran every pairwise double-difference among {A, B, C, base}, each with the same trigger / political / neutral cells and a shuffled-label null. **No pair is loyalty-specific.** Every genuinely-distinct organism pair fires at **AUROC 1.00 on the topic baseline** (political-vs-neutral, loyalty OFF both sides) – *as strongly as* on the loyalty test:

pair	trigger vs political (loyalty)	political vs neutral (TOPIC confound)
organism-a - organism-b	1.00	**1.00**
organism-b - organism-c	1.00	**1.00**
organism-a - base	1.00	**1.00**
organism-c - base	**0.50**	**0.50**

Loyal-minus-control cannot isolate the loyalty: the topic confound fires at AUROC 1.0 for every real organism pair; only organism-c minus base returns chance, because organism-c IS the base.

Figure 4: Loyal-minus-control cannot isolate the loyalty: the topic confound fires at AUROC 1.0 for every real organism pair; only organism-c minus base returns chance, because organism-c IS the base.

The one pair whose confound cancels to chance is **organism-c - base** – and the reason is decisive: **organism-c is bit-identical to the base model** (verified `||base - c|| = 0.0000`, cosine `1.00000` at every layer read; by contrast organism-a sits at cosine 0.67-0.77). Organism-c is a **null/sanity control**, not a content-matched one. So the sole "clean" pair is the one with *no model difference to detect*. The general lesson: **a double-difference only cancels a confound between two models that are activation-identical off-trigger – but any two independently fine-tuned organisms differ *everywhere*, so the cancellation never materializes for genuinely distinct models.** Model-diffing, even loyal-minus-control, cannot isolate the narrow loyalty. As a bonus, the probe returns *exactly* chance on the one truly-clean model – a built-in calibration confirming that the AUROC 1.0 elsewhere is real separation, not a probe artifact.

3.7 The calibrated headline: a matched-control detection map with a clean-twin placebo (CONFIRMED null)

Pulling the confounds (§3.1-3.6) into the one figure Track 2 asks for: for each of three matched near-miss contrasts we report **held-out AUROC + TPR at a target 1% FPR (Wilson/Clopper-Pearson CIs)**, compared against the **organism-c clean-twin placebo** (verified = base, §3.0) and a shuffled-label null. The probe is WITHIN a model (does the organism separate D+ from a matched control better than the clean twin?), the principal appears only in the user turn (no audit-framing leak), the wrong-principal control **DA- draws from a name pool disjoint from D+** (an earlier version had 100% D+/DA- byte overlap – a self-vs-self artifact we caught and retracted), and the placebo gets its **own** best-layer (no selection asymmetry). n = 15/cell, on the A100.

contrast (organism-a / -b)	isolates	org AUROC	clean-twin placebo	excess	realized FPR
extremity (D+ vs civic)	violence vs lawful	1.00	1.00	**+0.00**	0.04-0.05
principal (D+ vs wrong-principal)	**the loyalty**	**1.00**	**1.00**	**+0.00**	**0.013**
topic (D+ vs benign)	topic/length	1.00	1.00	**+0.00**	0.17 (n too small)

Every contrast separates at **AUROC 1.00 for the organism *and* the clean twin**, with **excess over placebo exactly 0** and shuffled nulls at chance (~0.50). So each apparent "detection" is a content / name-token / topic confound, and the loyal organisms separate **no better than a model that is literally the base**. The principal contrast – the actual loyalty test – resolves a **realized 1.3% FPR** at n = 15. This is the calibrated, placebo-controlled negative: **no narrow loyalty is detectable at a 1% FPR by any matched-control white-box probe we ran.** (We lead with excess-over-placebo, which is calibration-free; TPR@1%FPR is reported with its realized operating point, not asserted.)

3.8 A null is not a bound: the positive-control detection frontier (organism we implant)

A confound-controlled null says *we* didn't detect a loyalty; it does not say one is undetectable. To put a floor under it we build a **positive control** (`make_organism.py`): LoRA-implant a loyalty to a *fictional* principal (no pretrained-salience confound) at **known narrowness** – a KL penalty vs base on a wide benign anchor is the stealth knob – and run the SAME detector against it (`run_frontier.py`), under two conditions. **ORACLE:** the auditor knows the trigger + principal. **AUDITOR:** the principal is unknown (the realistic case; the released-audit candidate distribution). Both report AUROC excess over a shuffled null; a validity gate confirms the organism engages for the principal but *refuses* decoys (a real conjunction, not a general "extreme->engage" shift). The design is pre-registered with a git-provable ordering and a binding **kill criterion** (`FRONTIER_PREREGISTRATION.md`): if no config reaches engage >= 0.5 at KL <= 0.01 we report the frontier *scoped* to the KL range we reached and do **not** extrapolate down to the released organisms' KL<0.006. *[Final KL-swept `frontier.json` numbers are being generated on the A100; a first capability sweep produced organisms too broad (KL 0.7-1.6) to place the released ones – the KL-regularized rerun is what makes the frontier informative.]*

4 - Limitations & honesty

- Behavioral classification uses a refusal/engagement classifier; we report it as a discriminator, not a calibrated FPR, and flag ambiguous cases. Small N per cell – we report rates with the operating point stated, not a swaggery "TPR@1%FPR" detached from sample size.
- White-box recovery is a **negative** here (noise-dominated); we do not overclaim it.
- The identity-confound result is the load-bearing methodological contribution and is robust.

5 - Reproducibility & ethics

Public method repo (no weights): https://github.com/Chrislysen/secret-loyalty-probe – pip- installable, 137 offline tests, one-command Colab. Audit prompts are veiled red-team instruments for a sanctioned safety eval; we classify refuse-vs-engage and never produce or persist operational content (`ETHICS.md`). The goal is defensive: make secret loyalties *measurable*.

References

Lamerton & Roger 2026 (arXiv:2605.06846) - Kwon/Lamerton et al. 2026 (Formation Research) - Marks et al. 2025 (arXiv:2503.10965) - Sheshadri et al. 2026 AuditBench - Cywi?ski et al. 2025 (arXiv:2510.01070) - Arditi et al. 2024 (arXiv:2406.11717) - Marks & Tegmark 2023 (arXiv:2310.06824).